

Why Enterprises Pay the Frontier Premium

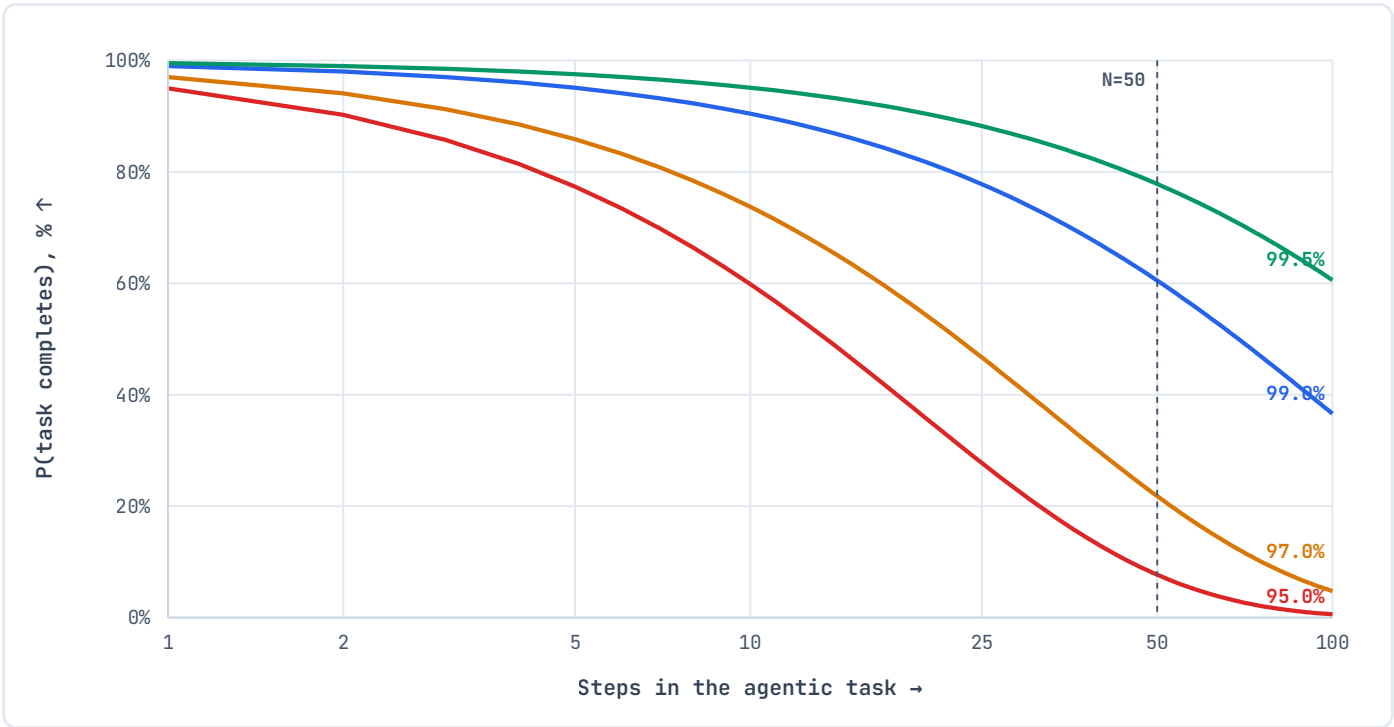
Four mechanisms behind the closed-vs-open procurement decision · modelled economics · August 2026

Token price is the wrong denominator. Model spend runs ~1-2% of loaded labour, value is convex rather than linear in capability, per-step reliability compounds as p^N over agentic horizons, and self-hosting only clears a cheap open-weights API above ~48% sustained utilisation – well above the 20-35% band real enterprise load runs at. The genuine overpayment isn't choosing proprietary; it's routing everything to a flagship max-effort tier.

1 · Agentic error compounding

Task completion probability vs horizon · $P(\text{success}) \approx p^N$

The largest single driver, and the one invisible on single-turn benchmarks. A two-point per-step reliability delta rounds away in a composite index and is **decisive** by step 50.



PER-STEP RELIABILITY	10 STEPS	25 STEPS	50 STEPS	100 STEPS
95.0%	59.9%	27.7%	7.7%	0.6%
97.0%	73.7%	46.7%	21.8%	4.8%
99.0%	90.4%	77.8%	60.5%	36.6%
99.5%	95.1%	88.2%	77.8%	60.6%

WHY THIS DOMINATES

Moving per-step reliability from 97% to 99% is a 2.8× swing in completion at N=50 and 7.4× at N=100 – the boundary between "the agent ships the PR" and "an engineer babysits and re-drives." No token saving recovers a workflow that doesn't complete.

2 • Self-hosting total cost of ownership

Effective cost per 1M output tokens vs fleet utilisation • $4 \times 8 \times \text{H200} + 3 \text{ ops FTE}$

Open weights are free; **serving them is not**. Fixed cost amortises over tokens actually produced, so unit cost is governed by utilisation – and enterprise diurnal load runs **5-10× peak-to-trough**.



BREAK-EVEN VS OPEN-WEIGHTS
3RD-PARTY API

47.9% util

BREAK-EVEN VS MID-TIER
PROPRIETARY

19.1% util

BREAK-EVEN VS FRONTIER
FLAGSHIP

3.8% util

FIXED COST MODELLED

\$1.81M /yr

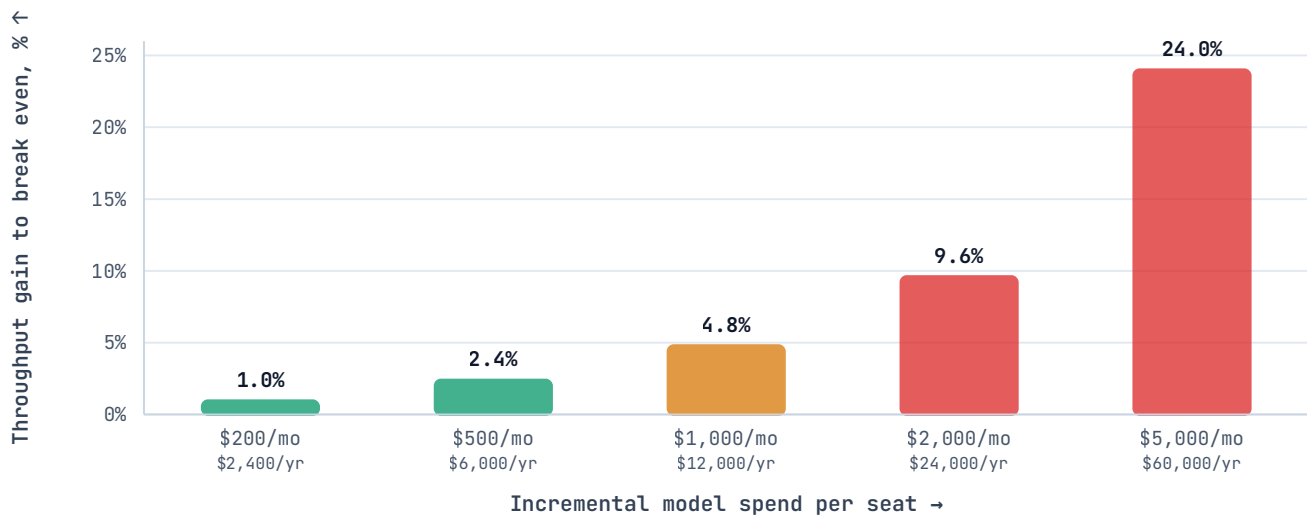
MODEL ASSUMPTIONS – LOAD-BEARING, TAGGED

Infra **\$911k/yr** (4 nodes × \$26/hr rented – MED) plus ops **\$900k/yr** (3 FTE × \$300k loaded – MED) = **\$1.81M/yr fixed**. Peak capacity 5,000 output tok/s per node for a ~100B-class MoE under heavy batching – LOW confidence; this is the swing variable. Halve throughput and every crossover doubles. The qualitative result survives that: self-hosting beats a flagship price almost immediately and a cheap open-weights API only at high sustained load.

3 • The labour denominator

Throughput gain required to repay the frontier premium • \$250k loaded labour per seat

The premium is repaid in productivity, not tokens – the question is what throughput lift the incremental spend has to buy.



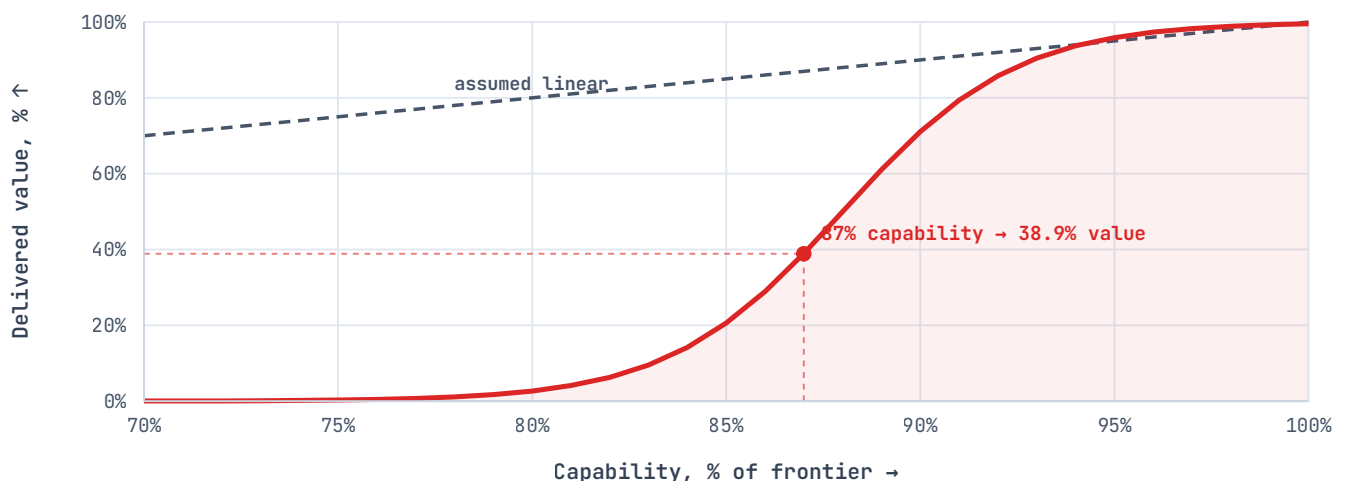
THE BAND WHERE THIS IS A NON-DECISION

At a realistic per-seat delta of **\$200-500/month**, break-even is a **1-2.4% throughput gain** – below measurement noise, which is exactly why enterprises stop optimising here. The argument only becomes live above **~\$2,000/seat/month**, where a **9.6%** lift is required and the claim needs evidence. Agentic workloads can reach that band, which is where routing discipline starts to pay.

4 • Threshold-gated value

Delivered value vs capability • the composite-average trap, priced

"Open weights reach ~87% of proprietary" assumes linear value transfer. On work gated by a pass threshold **delivered value is sigmoid**, and the last few index points sit on the knee.



READING IT

On this curve **87% of frontier capability delivers ~39% of the value**. The benchmark isn't wrong – a mean over ten evals is simply the wrong functional form for threshold-gated work. The easy 80% of tasks were automatable two model generations ago; the residue is exactly what sits on the knee. Illustrative – LOW confidence on curvature, HIGH on direction.

Net read

Eight drivers, with basis and confidence

DRIVER	MECHANISM	BASIS	CONF
Labor-dominated denominator	Model spend $\approx$ 1-2% of loaded engineer cost	Structural	HIGH
Threshold-gated value	87% composite capability $\rightarrow$ ~39% delivered value on gated tasks	Modelled	MED
Agentic error compounding	97% $\rightarrow$ 99% per-step = 2.8 $\times$ completion at N=50	Derived (p ^N)	HIGH
Calibration inversion	Opus 4.7 hallucination 36% vs GPT-5.5 xhigh 86%	AA, Jun 2026	HIGH
Self-host TCO	Break-even vs cheap API needs $\geq$ 48% sustained utilisation	Modelled	LOW
Indemnity & compliance	IP indemnity, SOC 2 / HIPAA / FedRAMP, residency	Contractual	HIGH
Provenance & sanctions	Training-data lineage, backdoor risk, Entity List exposure	Qualitative	MED
Harness, not checkpoint	Caching, batch, tool-call reliability, agentic scaffolds	Qualitative	HIGH

LANE 1 • OPEN WEIGHTS

Bounded, high-volume, private

Classification, extraction, routing, embedding. Error-tolerant, short-horizon, sustained enough to clear the utilisation bar. **The open-weights case holds cleanly here.**

LANE 2 • MID-TIER PROPRIETARY

General reasoning, good-enough

Where the **dominated-flagship trap** lives. Cost-tuned proprietary out-values open weights on both axes – same index, a fifth the cost.

LANE 3 • FRONTIER

Long-horizon, calibration-critical

Agentic chains, must-not-hallucinate, threshold-gated. p^N and the sigmoid knee are both load-bearing. **Token cost is the small line item.**

NET

Enterprises don't pay the premium because they're bad at arithmetic – they pay it because the arithmetic they're doing has a different denominator. Where open weights genuinely win is a real and sizeable quadrant, but it's **bounded work at sustained load**, not general reasoning. The overpayment that actually exists is **failure to segment traffic**, and it shows up inside the proprietary cohort as often as across the closed/open line.

Sources & confidence. Reliability compounding is derived arithmetic (p^N) – HIGH. Hallucination rates (Opus 4.7 36%, GPT-5.5 xhigh 86%, Gemini 3.1 Pro 50%) and index/cost figures from Artificial Analysis Intelligence Index v4.0, live June 9 2026 – HIGH. Self-host TCO is a **model, not a measurement**: node rental \$26/hr and \$300k loaded FTE are market estimates (MED); 5,000 output tok/s per node is an assumption (LOW) and dominates the result. Loaded labour \$250k is a representative US senior-engineer figure (MED). The threshold curve is illustrative – LOW on curvature, HIGH on direction. Reference API prices are per 1M output tokens and exclude input, caching, and the verbosity tax on output-heavy models, which shifts effective cost further.

Not included: project-level discounts, committed-use pricing, and enterprise volume agreements materially compress the frontier premium and are absent from every figure here.